

- 3 The five bases found in nucleic acids are described as nitrogenous organic compounds. There are two types of base.

Fig. 3.1 shows the structure of the five bases.

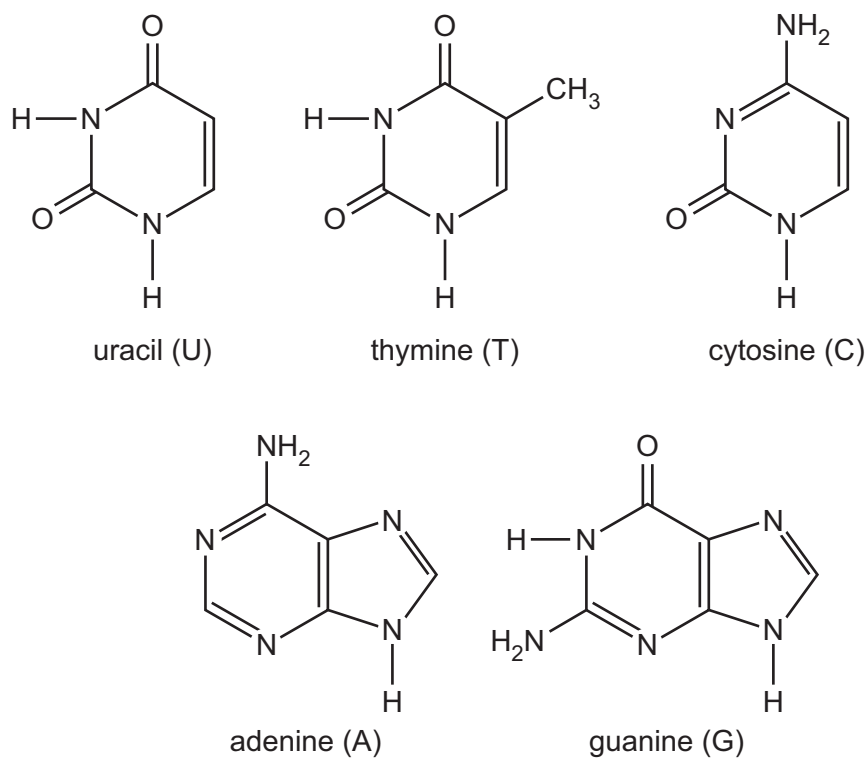


Fig. 3.1

- (a) (i) State the name of the type of base that includes U, T and C.

..... [1]

- (ii) State why it is **not** correct to say that all nucleic acids have five bases.

.....
..... [1]

(b) Fig. 3.2 shows a stage in the replication of DNA. The circled part is enlarged in Fig. 3.3 to show the elongation of the DNA strand that is being synthesised.

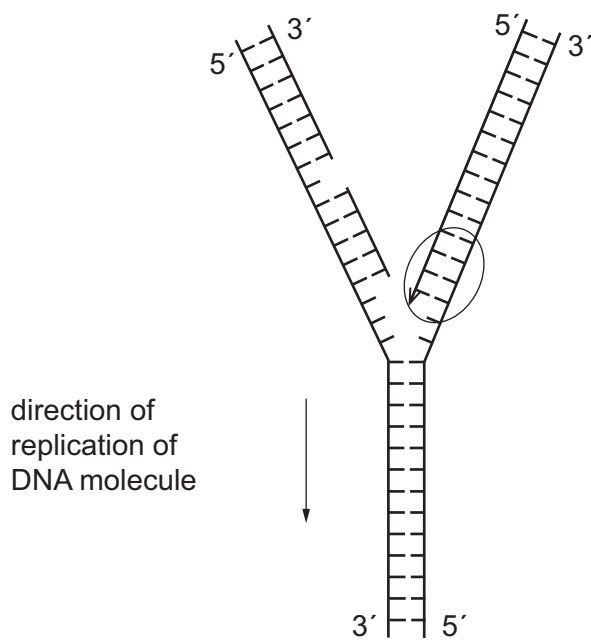


Fig. 3.2

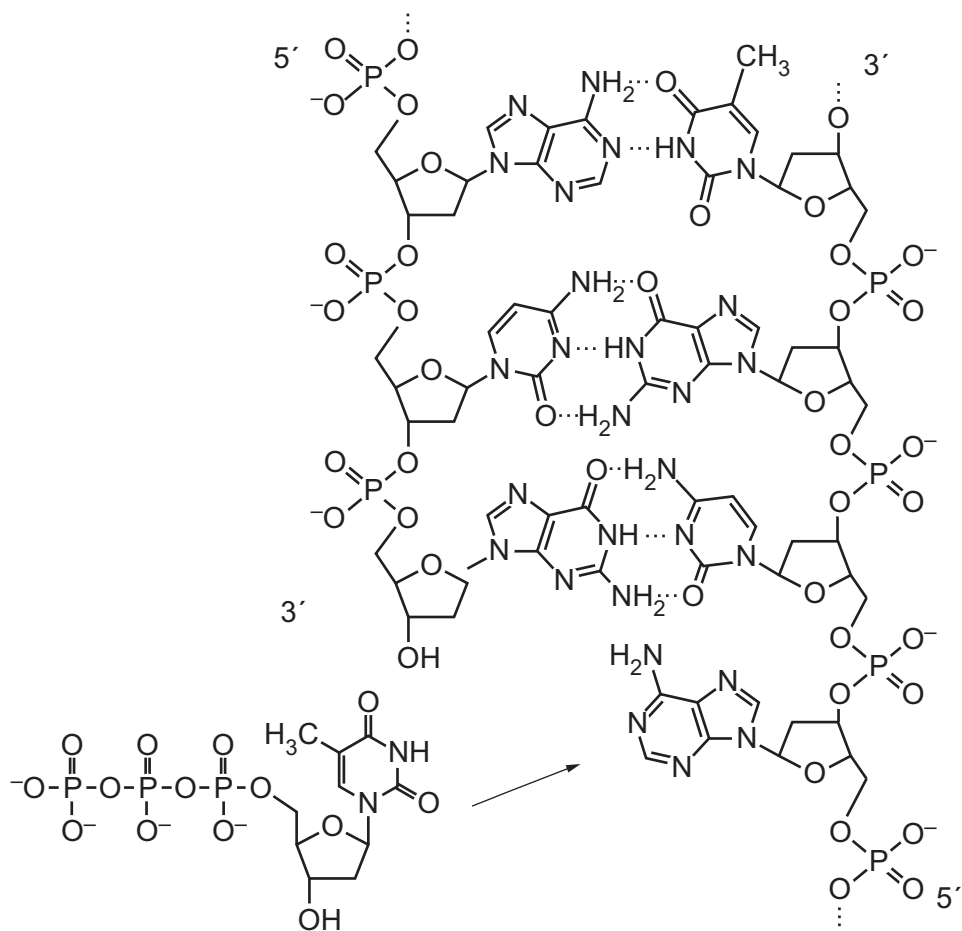


Fig. 3.3



Describe the sequence of events that occurs at the stage shown in Fig. 3.3 to extend the synthesised strand.

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

(c) The two strands in a molecule of DNA are described as antiparallel.

With reference to Fig. 3.2 and Fig. 3.3, state what is meant by antiparallel, **and** explain how the antiparallel arrangement of the strands determines how new strands are synthesised.

.....

.....

.....

.....

.....

.....

.....

..... [3]

[Total: 9]